



Financovaný
Európskou úniou
NextGenerationEU

PLÁN [OBNOVY]



ÚRAD PODPREDESEDU VLÁDY
SLOVENSKEJ REPUBLIKY
PRE PLÁN OBNOVY
A ZNALOSTNÚ EKONOMIKU



VÝSKUMNÁ
AGENTÚRA

SAMQNN · WORK PACKAGE WP4 · DELIVERABLE D15

A statistical approach to monitoring quantisation in neural networks

Monitoring of quantisation during training via a statistical similarity index of the weight distribution

Ing. Roman Budjač, PhD.

Project 09I03-03-V04-00562 · Recovery and Resilience Plan of the Slovak Republic · Component 9, Excellent science

1 Context and goal

- Quantisation reduces the memory and compute requirements of a model by representing weights in lower bit precision than the standard 32-bit format.
- This saving may come at the cost of a drop in accuracy, so it is important to be able to track the quality of quantisation and its impact.
- Our approach: monitor the extent to which the weight distribution of the quantised model deviates from the full-precision (FP32) model.
- WP3 implemented the approach as a new training strategy; WP4 verifies it experimentally.

4×

compression at 8 bits

≈ 0

accuracy loss (8-bit)

Why statistically?

- The weight distribution is exactly what quantisation changes.
- The index can be computed continuously during training, cheaply.
- It acts as an early-warning signal, not merely an after-the-fact indicator.

2 Proposed method

- A similarity index between the reference FP32 weight distribution and the current distribution of the quantised model.
- Primary method: Jensen-Shannon divergence: symmetric, bounded to $[0,1]$, suitable for thresholding.
- It is complemented by Kullback-Leibler divergence and cosine similarity (the measures are strongly correlated).
- Histograms on pre-fixed bin edges $\rightarrow$ a consistent comparison throughout training.
- Per-channel quantisation, the index globally and per layer.
- It is computed continuously through a custom Keras callback at the end of each epoch.

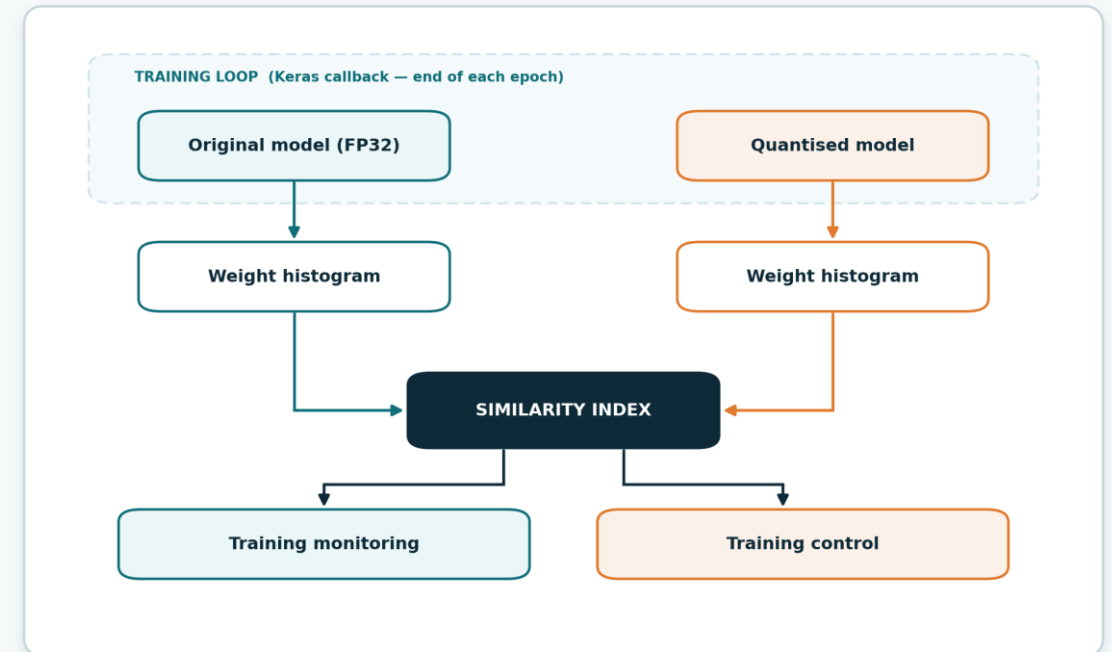


Fig. 1 Similarity index computed continuously during training through a Keras callback.

3 Verification approach

Datasets

digits · Fashion-MNIST · MNIST · CIFAR-10 · ImageNet-V2 validation subset

Architectures

small convolutional network → pretrained MobileNetV2 and ResNet50

Metrics

top-1 accuracy (also top-5 for ImageNet-V2) and the loss function vs FP32

4 criteria from D10

index threshold · correlation with accuracy · stability across epochs · reproducibility

4 The index faithfully tracks quantisation

-0.98 / -1.00

$\rho(\text{bits}, \text{JS})$: monotonic relationship

8-bit

practically lossless (including ResNet50)

2-bit

accuracy collapse

Main observations

- The Spearman correlation between bit depth and JS divergence is ≈ -0.99 across datasets (-1.00 on ImageNet).
- At 8 bits accuracy remains practically unchanged at fourfold compression, including the large models.
- During quantisation-aware training the index is stable, the weights remain genuinely quantised.
- Per-layer analysis: the first convolutional and the output layer are the most sensitive.

5 Compression and accuracy trade-off

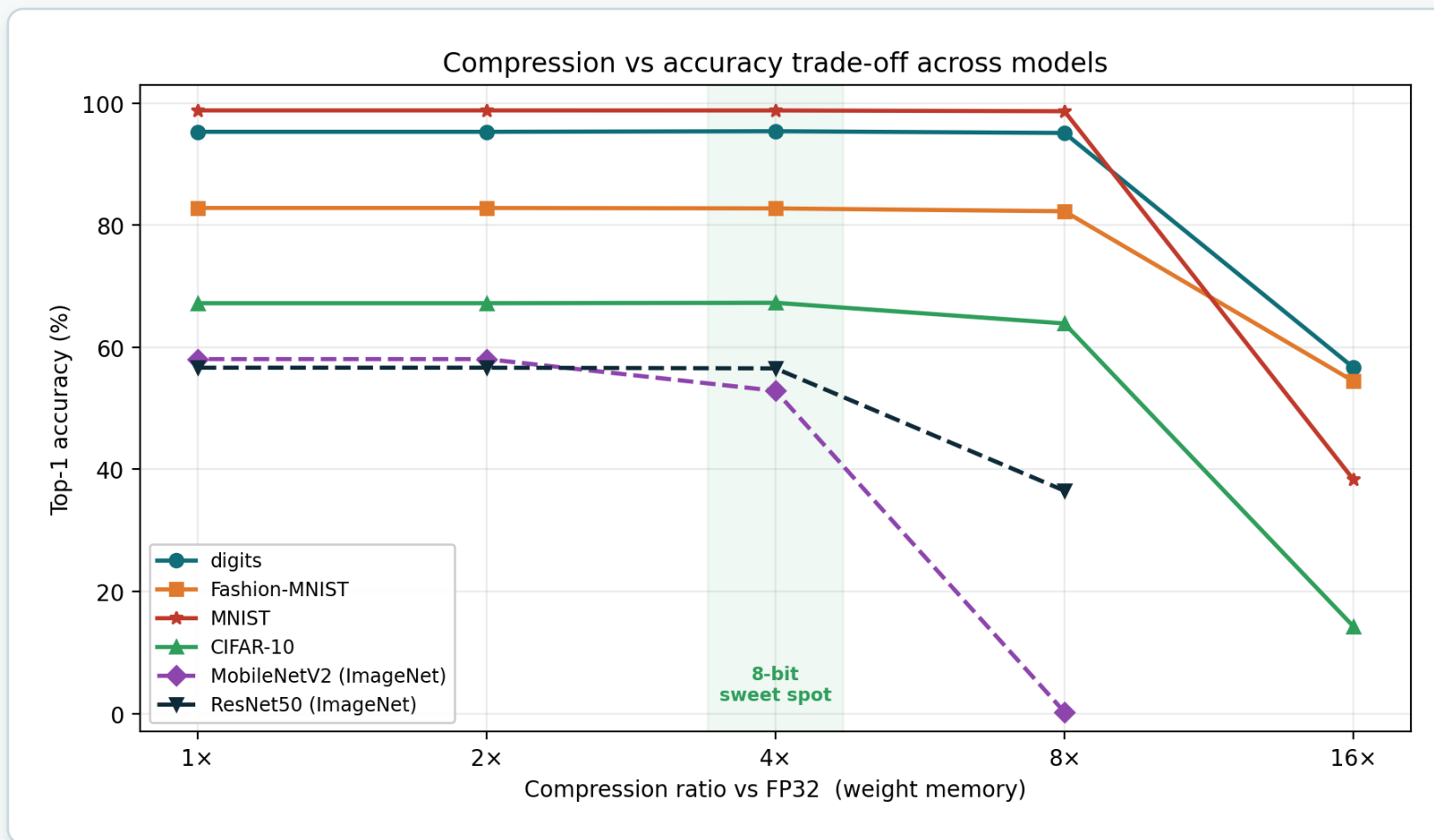


Fig. 2 X axis: weight-memory compression ratio (32/bits, log). Y axis: absolute top-1 accuracy. The band around 8 bits is the "sweet spot".

The band around fourfold compression (8 bits) is almost lossless across the models.

6 The index as an early-warning signal

Bit depth	Average top-1 drop	Loss increase (×FP32)	Average index (JS)	Compression
8	0.0%	1.00×	0.0001	4×
4	-1.7%	1.11×	0.0819	8×
2	-63.5%	20.33×	0.6322	16×

- Down to 8 bits the index and accuracy remain unchanged, fourfold compression almost for free.
- At 4 bits the index rises noticeably while accuracy drops only slightly ($\approx 1.7\%$), the index rises ahead of degradation.
- At 2 bits a collapse occurs. It is exactly this nonlinear lead that makes the index an early-warning signal.

7 Validation of the index

- The optimal bit depth is not determined by the index, it follows from direct accuracy measurement and established practice (8 bits $\approx$ lossless).
- This independently known fact serves as a criterion for validating the index.
- The index stays below the threshold exactly where accuracy is preserved (≥ 8 bits), and rises where it drops (≤ 4 bits).
- Agreement with the reference behaviour confirms that the index is a usable proxy indicator of quantisation quality.

⚠ Limit of validity

- For MobileNetV2 (4-bit) the weight index does not capture the accuracy collapse.
- The degradation originates outside the weight distribution, at the level of activations.
- The index is reliable for the tested family of models; both the threshold and the expressiveness (weights vs activations) must be calibrated for a given architecture.

8 Choosing the bit depth: 6 vs 8 bits

6 bits as analytical optimum

- When minimising bits, the strategy naturally chooses ≈ 6 bits.
- The index is well below the threshold ($JS \approx 0.001$), accuracy preserved.
- Almost lossless across the tested datasets.
- The most economical choice was a compression of 5.33 $\times$.

8 bits as the practical standard

- The recommended target for deployment (INT8).
- Integer hardware supports it natively: mobile and edge NPUs, INT8 accelerators, TFLite/ONNX kernels.
- 6 bits is not used in HW-native form (i.e. in practice it would be stored in 8 bits anyway).
- The method offers both: an optimum and a deployable standard.

9 Strategy: SMT vs baseline

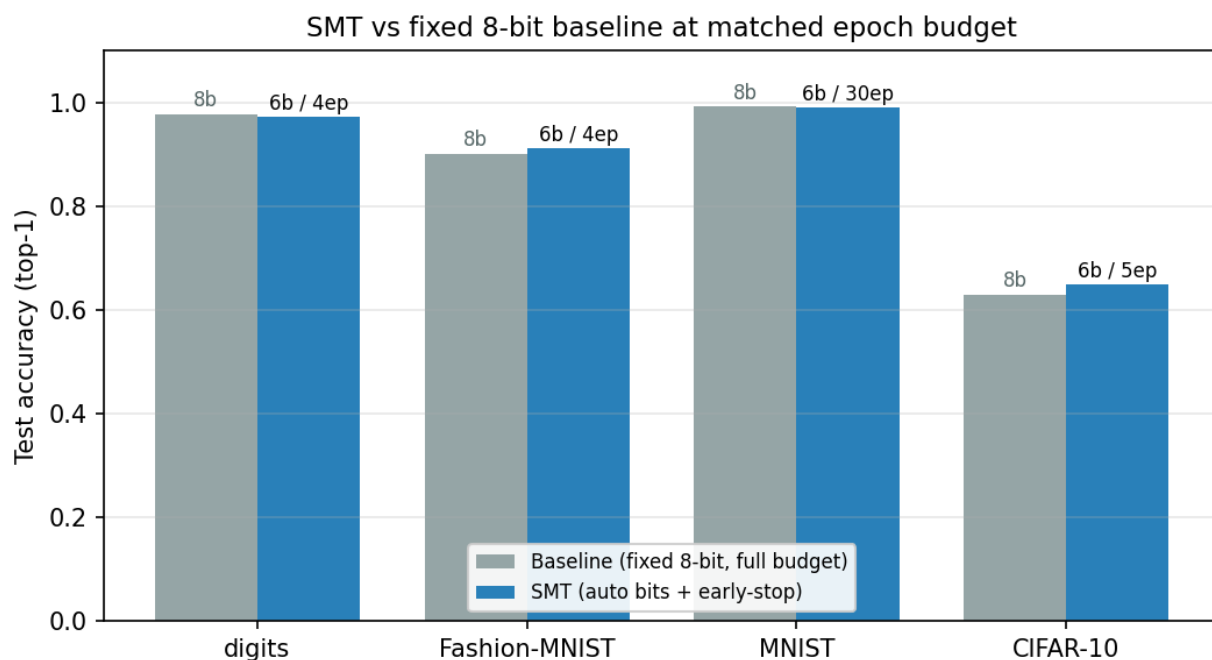


Fig. 3 SMT (auto ≈ 6 bits + early stopping) vs a fixed 8-bit baseline at the same epoch ceiling.

- For each dataset SMT automatically chooses the bit depth according to the index threshold and tracks the per-layer index.
- Across all 4 datasets it chooses ≈ 6 bits ($5.33\times$), outperforming the 8-bit baseline ($4\times$) in memory savings.
- Accuracy remains comparable to the baseline (differences within noise), 6 bits is practically lossless.
- Early stopping triggered on 3 of 4 datasets $\rightarrow$ a significant saving of training epochs.

10 Early stopping as a detector of index stabilisation

- The mechanism stops training when the per-layer index stabilises, when the weight distribution stops changing.
- Safety property: it triggers only on stabilisation, so it never stops prematurely. If it does not trigger, training simply runs to the ceiling.
- On the full run the index stabilised quickly and saved a substantial part of training without loss of accuracy.
- We report the stabilisation epoch as a measure of how quickly quantisation stability is reached, not as a failure.

4-5

epochs (digits, Fashion,
CIFAR)

30

MNIST, ran to the ceiling

Recommended use

- Together with an epoch ceiling.
- Together with an accuracy check.
- Not as the sole stopping criterion.

11 Benefits and limitations

Benefits

- Computationally cheap, overhead negligible compared with a single pass of the network.
- Dataset agnostic and independent of the choice of metric.
- Two modes: predictor (bit selection before training) and monitor (during training, including early stopping).
- Per-channel quantisation → robust on deep networks.

Limitations

- The weight index does not capture degradation outside the weight distribution (MobileNetV2 4-bit → activations).
- The threshold value is architecture dependent, it must be calibrated.
- It is recommended to track the index together with accuracy.
- ImageNet verified on the V2 validation subset.

Conclusion and future directions

The statistical similarity index is a usable and low-cost tool for monitoring quantisation directly during training. It satisfies the D10 criteria and fulfils Objective 6, the concept proven on standard datasets, evaluated against FP32 and the D10 criteria.

Future directions

- Extending monitoring from weights to activations as well.
- A mixed-precision strategy driven by the per-layer index.
- More thorough calibration of the threshold values.

The results form the basis for the upcoming scientific publication (D16) and the final project report (D17).